Exact Periodic Orbits and Algebraic Zeta Geometry of One-Vertex Edge-Type Dyck Shifts

HCS Research Program

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Abstract

We close the periodic-point theory of the edge-type Dyck shift associated with one vertex and N loop edges. The algebraic zeta function yields explicit odd and even fixed-count formulas for every N and every period, followed by primitive points and orbits through Möbius inversion. We distinguish the $N = 1$ full-two-shift boundary from the $N > 1$ double dominant pole, quadratic branchpoints, nonrationality, and orbit asymptotics. Formal series, symbolic algebra, and direct periodic-word enumeration independently audit the result; the source-locked entropy is $\log(N + 1)$.

1 Model and algebraic zeta

Let D_N^E be the edge-type Dyck shift for the graph with one vertex and N loop edges. Its alphabet is $\{a_i, b_i : 1 \le i \le N\}$, with inverse-monoid relation $a_i b_j = 1$ for $i = j$ and 0 otherwise. A finite block is admissible when its reduction is nonzero. Put

$$s_N(z) = \sqrt{1 - 4Nz^2}.$$

The context-free circular-code series satisfies

$$g_N(z) = \frac{Nz^2}{1 - g_N(z)}, \qquad g_N(z) = \frac{1 - s_N(z)}{2},$$

where the second expression is the small solution. The source theorem, specialized to the one-vertex N-loop graph, gives

$$\zeta_N(z) = \frac{1 - g_N(z)}{(1 - Nz - g_N(z))^2}.$$

Substitution, rather than a claim of priority, yields

$$\zeta_N(z) = \frac{2(1 + s_N(z))}{(1 + s_N(z) - 2Nz)^2}. \tag{1}$$

Theorem 1 (all fixed and primitive counts)**.** *Let $F_N(n) = |\operatorname{Fix}(\sigma^n)|$. If n is odd, then*

$$F_N(n) = 2\left((N + 1)^n - \sum_{j=0}^{(n-1)/2} \binom{n}{j} N^j\right).$$

If n is even, then

$$F_N(n) = 2\left((N + 1)^n - \sum_{j=0}^{n/2} \binom{n}{j} N^j\right) + \binom{n}{n/2} N^{n/2}.$$

The numbers of least-period points and primitive orbits are

$$P_N(n) = \sum_{d|n} \mu(n/d) F_N(d), \qquad C_N(n) = P_N(n)/n.$$

Proof. By definition, $z\,\zeta_N'(z)/\zeta_N(z) = \sum_{n\ge 1} F_N(n) z^n$. Insert (1), expand the quadratic series $s_N(z)$, and separate the central binomial term when n is even. The two displayed parity formulas result. Möbius inversion isolates least periods. A least-period-n orbit has exactly n possible origins, proving the last division. □

2 The origin-marked word convention

$F_N(n)$ counts points of $\mathrm{Fix}(\sigma^n)$, equivalently length-n periodic words with a distinguished origin; it does not count cyclic necklaces. Our direct audit enumerates such words and checks every cyclic factor through length $2n$ in the periodic extension. A stack reduction cancels $a_i b_i$ and rejects $a_i b_j$ for $i \ne j$. Thirty-three (N, n) cells, including $N = 1, \dots, 6$, agree exactly with Theorem 1. Only after Möbius inversion is $P_N(n)$ divided by n.

This distinction is substantive: division of $F_N(n)$ before inversion would mix points whose least period properly divides n and generally fail integrality.

Lemma 1 (two-period sufficiency)**.** *An n-periodic extension is admissible if and only if every cyclic factor of length at most $2n$ has nonzero reduction.*

Proof. Every nonzero reduced word has normal form BA, with unmatched closing letters B followed by unmatched opening letters A. Fix a cyclic origin and call its period word v. If v reduces to zero, a forbidden prefix already has length at most n. Otherwise, in v^2 the only new interaction is the AB interface. A colour mismatch is therefore already visible there. If the interface is compatible, equal lengths cancel and unequal lengths leave a monotone excess on one side; crucially, the final copy of A is unchanged. Every later copy, and every prefix of it, repeats the same interface comparison. Thus the first zero prefix has length at most $2n$. Auditing every cyclic origin covers every factor of the bi-infinite extension. □

3 Boundary, poles, and asymptotics

Theorem 2 (singularity dichotomy)**.** *For $N = 1$, the radical in* (1) *cancels and $\zeta_1(z) = (1-2z)^{-1}$, so $F_1(n) = 2^n$. For $N > 1$, the point $r_N = (N+1)^{-1}$ is a double pole, strictly inside the quadratic branchpoints $\pm(2\sqrt{N})^{-1}$. The zeta function is nonrational and*

$$F_N(n) = 2(N+1)^n + O\!\left(\frac{(2\sqrt{N})^n}{\sqrt{n}}\right), \qquad C_N(n) \sim \frac{2(N+1)^n}{n}.$$

Moreover, $h_{\mathrm{top}}(D_N^E) = \log(N+1)$.

Proof. At r_N, $s_N(r_N) = (N-1)/(N+1)$, so the squared denominator vanishes while the numerator and the derivative of its unsquared factor do not. Since $N + 1 > 2\sqrt{N}$ for $N > 1$, this pole precedes both branchpoints. Replacing s_N by $-s_N$ changes (1); quadratic conjugation therefore rules out rationality. The omitted binomial tail is bounded by a geometric multiple of its central term, which is $O((2\sqrt{N})^n/\sqrt{n})$. Proper divisors are exponentially smaller, and Möbius inversion gives the cycle asymptotic. Krieger–Matsumoto Proposition 3.1 proves for Markov-Dyck shifts that topological entropy equals the exponential periodic-point growth rate. The displayed asymptotic therefore gives $h_{\mathrm{top}}(D_N^E) = \log(N+1)$. This uses that source theorem, not an unproved general identification of periodic growth with entropy. □

4 Exact audit and scope

For $N = 1, \dots, 6$ and $n = 1, \dots, 24$, all 144 counts from the binomial formula equal an independent exact formal logarithmic derivative. SymPy checks 72 coefficients plus cancellation, pole, dominance, conjugation, and six entropy/dominant-radius identities. Byte replay is exact. Eighteen semantic attacks with repaired payload hashes and one separate stale-hash integrity attack are rejected.

$N\backslash n$	1	2	3	4	5	6
1	2	4	8	16	32	64
2	4	12	40	120	384	1152
3	6	24	108	432	1836	7344
4	8	40	224	1120	5888	29440

This algebraic zeta is not identified with a target local factor. We claim no root numbers, automorphy, target determinant, or Hilbert–Pólya operator. The registered tuple is

`(A0_FAIL, A1_WEAK, A2_FAIL, A3_FAIL, A4_FAIL).`

The exact record is `overall=ROUTE_A_REJECTED` and `route_b_invocation_allowed=false`.

Source note

The model, zeta formula, and entropy lock follow Krieger–Matsumoto [1]; Keller's loop-counting framework gives related context [2]. The official Münster volume PDF controls pagination; an arXiv journal-reference entry reports 171–185. No priority claim is made.

References

[1] W. Krieger and K. Matsumoto, Zeta functions and topological entropy of the Markov-Dyck shifts, *Münster J. Math.* 4 (2011), 171–184, arXiv:0706.3262.

[2] G. Keller, Circular codes, loop counting, and zeta-functions, *J. Combin. Theory Ser. A* 56 (1991), 75–83, doi:10.1016/0097-3165(91)90023-A.

Declarations

Scope. The exact registered literal is `NO_BAD_EULER_OR_ROOT_NUMBER`. **Data and code.** Exact ledgers, independent checks, and build instructions are supplied in the HCS-C205 release package. **Competing interests.** None declared. **AI-use disclosure.** Generative AI tools assisted drafting and code generation. Every theorem statement, source record, exact output, and scope claim was checked through the released internal artifact audit chain; this is not external peer review or an independent error process.